Ice Rumble

WORKING TITLE

A LAN party game for 2–4 players: short physics mini-games, one shared scoreboard.



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Concept render, AI-generated. Not gameplay.

How it plays: one room, short rounds, one scoreboard

2–4 friends in one room, each at their own PC. One PC hosts; the others join over the local network. **Everyone plays the same cube, in one of four colours.**

1 Lobby

Host starts a session. Others join with the host's local IP.

2 Round

Host picks the next mini-game. Everyone loads the same arena.

3 Play

Host runs the physics. Clients send input and draw the result.

4 Place

Round ends. Placement becomes points.

5 Standings

Scoreboard updates. After the last round: overall winner.

NEXT MINI-GAME

PLACEMENT POINTS · PROPOSED

1st	4 PTS
2nd	3 PTS
3rd	2 PTS
4th	1 PTS

BETWEEN ROUNDS

CARRIES OVER	Points. Nothing else accumulates.
CATCH-UP	The last round's winner earns slightly fewer points next round; last place earns slightly more.
RESETS	Mass and position, so one bad round does not snowball.

One value is your health, your ammunition and your weight



A cube spits out a piece of its own body. Every shot makes it smaller, hit or miss.

SHOOTER MASS

Faded part: lost in this exchange



The target loses mass on impact and is pushed back along the line of the shot.

TARGET MASS

Faded part: lost in this exchange



Less mass means the same hit pushes further. Off the edge is a knockout.

KNOCKBACK, SAME HIT

Grows as mass drops

Shooting, getting hit and getting launched all read from one number the player can see: **the cube's size.**

Bars show direction only. Costs and knockback values are set in playtests.

Why both the shooter and the target lose mass

VARIANT A

Only the shooter pays



BEST PLAY

Shooting is pure loss. The best play is to never shoot and wait for the others to weaken each other.

✕ REWARDS HIDING

VARIANT B

Only the target pays



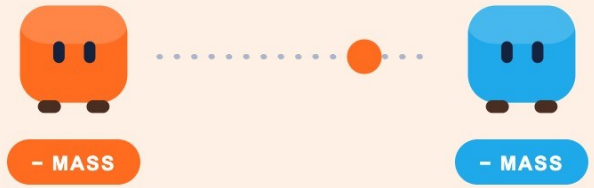
BEST PLAY

Shooting is free. The best play is to shoot nonstop in every direction.

✕ REWARDS SPAM

VARIANT C

Both pay: our rule



BEST PLAY

Every shot is a trade. A hit should take more from the target than the shot cost you, and a miss costs only you.

✓ REWARDS AIM AND TIMING

A shot only pays off when it lands, so **accuracy becomes the skill.** And never shooting scores nothing: points come from knockouts, not survival.

Exact costs are tuning values. We fix them in playtests, not on paper.



FIG 2 · Orange spits a piece of itself; Cyan is hit near the edge. Concept render, AI-generated.

MINI-GAME 01

Mass Knockout

GOAL

Knock other cubes off the floating platform.

CONTROLS

Move, aim, spit. A short cooldown between shots.

SCORING

A knockout goes to the last cube that hit the one that fell. Most knockouts at the timer wins.

RESPAWN

After a short delay, at full mass.

ARENA

One flat square platform. Open edges, no walls.

MINI-GAME 02

Ice Race

GOAL

Finish the laps first on a floating ice track.

CONTROLS

Steering changes acceleration, not velocity. Braking restores grip.

SKILL

Low sideways grip: carry speed through corners without sliding off.

FALLS

Respawn at the last checkpoint. You lose time, not the race.

PLACEMENT

Finish order. Unfinished players ranked by track progress.



FIG 3 · Racing a floating ice track; the HUD is part of the render. Concept render, AI-generated.

One shared core; each mini-game plugs into it

MINI-GAMES

Mass Knockout
mass state · spit · knockback

Ice Race
ice profile · track · checkpoints

Next mini-game
new rules + one arena

MINI-GAME CONTRACT

Setup(players) → Run() → Results() : placement

SHARED GAMEPLAY

Character | **Camera** | **Round state machine** | **Scoreboard** | **Lobby & UI**

NETWORK

Host-authoritative LAN · Unity Netcode for GameObjects (candidate)

ENGINE

Unity · built-in rigidbody physics

WHY THIS SPLIT

Reuse

A new mini-game is new rules plus one arena. Character, camera, network and scores already exist.

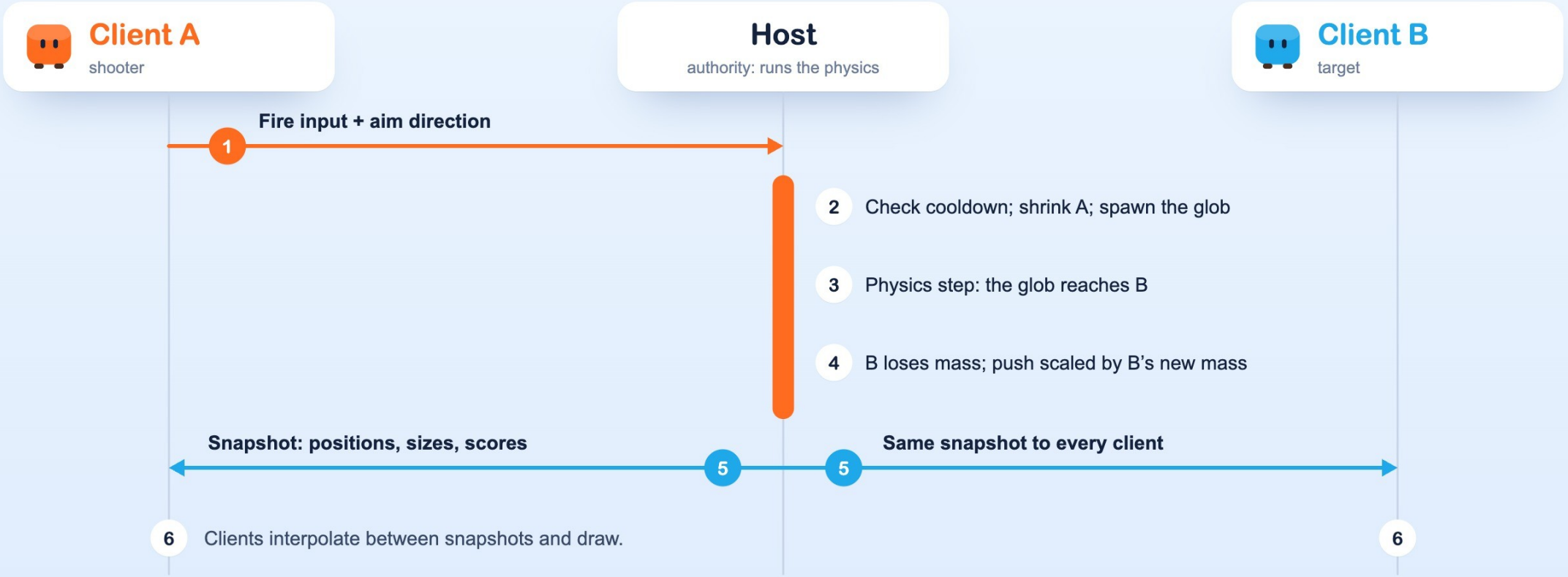
Per-game physics

Each mini-game sets its own profile: low grip on ice, firm grip in Knockout.

One truth

Everything that decides a result runs on the host, so every screen agrees.

One shot over the LAN, step by step



Your own movement is predicted locally, so it feels instant. Hits, mass changes and knockouts are decided only on the host. **Main risk:** keeping physics smooth on clients. LAN latency is low, and we tune the snapshot rate.

Social PC play is growing; LAN party games are rare

$43.6B

PC games revenue in 2025, up 12% year on year.

Newzoo [1]

$4.1B

Steam revenue of games with co-op in H1 2025, an all-time high (estimate).

Alinea [2]

70%

of US players aged 8+ play with other people.

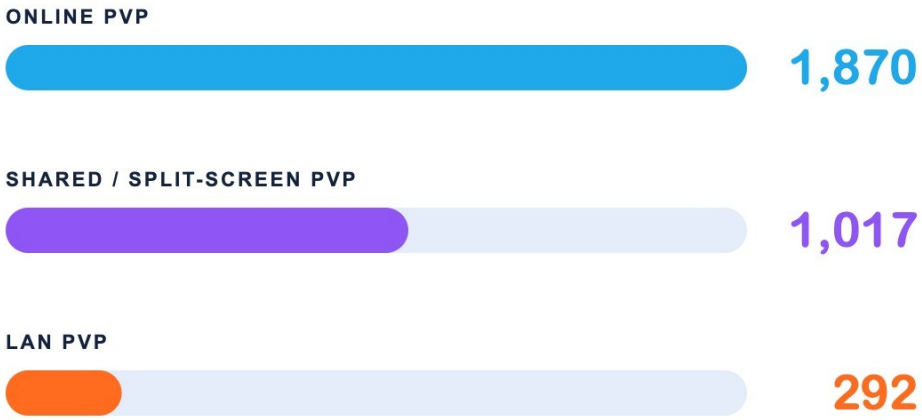
ESA 2026 [3]

≈8%

of ~3,490 Steam “Party Game” titles list LAN PvP.

Steam, Oct 2026 [4]

STEAM “PARTY GAME” TITLES BY MULTIPLAYER TAG



Steam store search, Oct 2026 [4]. Tags are user-applied.

RISK, STATED PLAINLY

An estimated 65.9% of 2025 Steam releases earned under $1,000 [5], while 5,863 games passed $100K in 2025 (Valve) [6]. **LAN is our differentiator, not a demand driver.**

Five games we are measured against



Party Animals

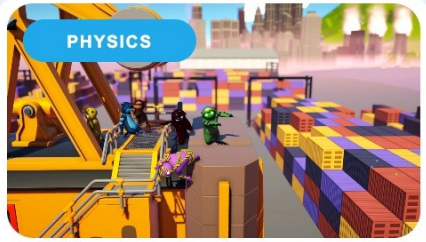
Recreate Games · 2023

VISUAL
Plush 3D animals, toy-like arenas.

MECHANIC
Physics brawling: grab, punch, throw rivals out.

MULTIPLAYER
Online and same-screen. No LAN listed.

FINANCIALS
Launch peak 104,174 players. Owner est. 2.5M–6.0M.



Gang Beasts

Boneloaf · 2017

VISUAL
Soft, gelatinous 3D people.

MECHANIC
Ragdoll brawling into stage hazards.

MULTIPLAYER
Same-screen and online. No LAN listed.

FINANCIALS
85% positive. Owner est. 4.6M–5.8M.



Fall Guys

Mediatonic / Epic · 2020

VISUAL
Candy-coloured beans, obstacle courses.

MECHANIC
Short race and survival rounds in a playlist.

MULTIPLAYER
Online, cross-platform only.

FINANCIALS
Free-to-play. Legacy Steam peak 172,213. Metacritic 80.



Pummel Party

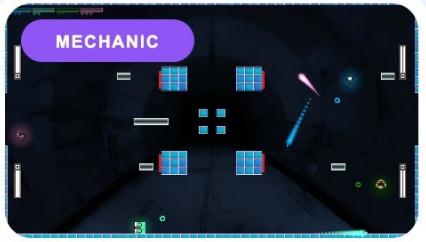
Rebuilt Games · 2018

VISUAL
Colourful low-poly 3D, slapstick items.

MECHANIC
Board game with short competitive mini-games.

MULTIPLAYER
4–8 online and local. No LAN listed.

FINANCIALS
$14.99. Peak 20,957. Owner est. 1.5M–7.6M.



Squarena

Codeaclysm · 2022

VISUAL
Minimal flat 2D squares.

MECHANIC
Arena shooter: low health shrinks you; small bodies feel more recoil.

MULTIPLAYER
Up to 6 local. No LAN listed.

FINANCIALS
$4.99. Sales not disclosed.

Screenshots: official Steam media, © publishers. “Owner est.” = third-party estimates via SteamDB, not sales. 2026-10-17 [7].

Where we sit among them

	PARTY ANIMALS	GANG BEASTS	FALL GUYS	PUMMEL PARTY	SQUARENA	ICE RUMBLE
3D characters	✓	✓	✓	✓	○	✓
Ranged attacks	–	○	○	–	✓	✓
Damage changes your body size	○	○	○	○	✓	✓
Less mass, further knockback	○	○	○	○	○	✓
Several mini-games, one result	○	○	–	✓	○	✓
Official LAN play	○	○	○	○	○	✓
Plays with no internet	○	✓	○	✓	✓	✓

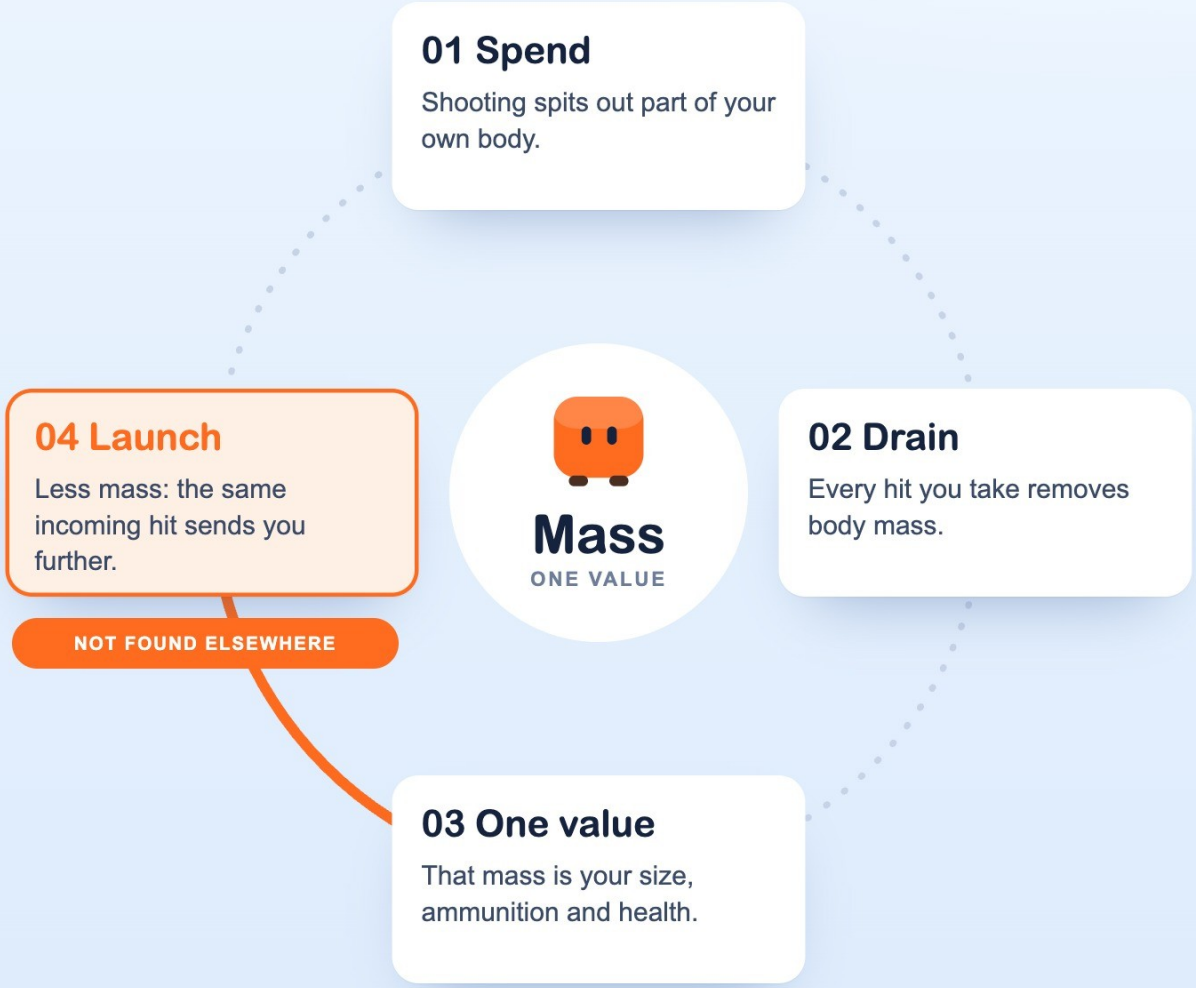
✓ Yes – Partly ○ No ✓ Planned (Ice Rumble)

“Official LAN: No” = not advertised in the game's Steam features. “Partly”: Party Animals and Pummel Party have some weapons or items; Fall Guys' playlist eliminates rather than scores. Party Animals needs internet even on one screen [8]. Ice Rumble column = planned design.

THE NEW LINK

No shipped game we found ties knockback to the mass you have lost.

Mass Knockout puts four links on one physical value the player can see. Links 01 to 03 have relatives in other games. **Link 04 we did not find.**



Prior-art search of shipped games, Oct 2026 [9–12]. Evidence on the next slide.

Closest prior art for each link, and what we did not find

LINK		CLOSEST PRIOR ART	STATUS
01	**Spend your body**	Osmos ejects its own matter to move, not to attack. Agar.io ejects mass.	DIFFERENT USE
02	**Hits shrink you**	Squarena (2D): lower health, smaller body. Morphies Law: hits move mass between robots.	PRECEDENT
03	**Attack costs health**	Crisol: Theater of Idols: your blood is your ammunition. Not physical mass.	PRECEDENT
04	**Lighter, launched further**	Smash Bros. raises knockback with a damage % counter, not mass. In Squarena, smaller bodies feel more of their own recoil.	NOT FOUND
01–04	**All four in one 3D party game**	No shipped game found.	NOT FOUND

Prior-art search of shipped games, Oct 2026 [9–12]. “Not found” means not found after searching.

Same rules, different builds: weight is also ammunition



FIG 4 · Heavy, balanced and light builds of the same character. Concept render, AI-generated.

Weight classes are standard in the genre (e.g. Mario Kart). The twist: here weight is also your ammunition, so a build changes how you play, not only how you look.

Heavy

More starting mass: more shots to spend, harder to push. Moves slower.

MASS · AMMO

SPEED

Balanced

The reference build every other one is tuned against.

MASS · AMMO

SPEED

Light

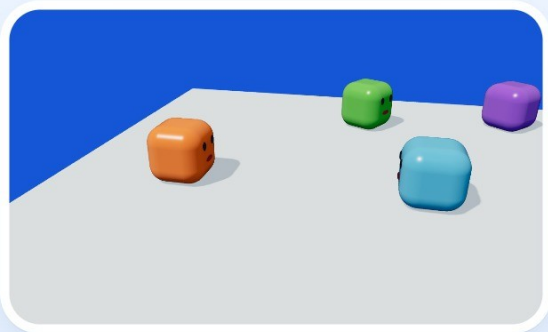
Less starting mass: fewer shots, flies further when hit. Moves faster.

MASS · AMMO

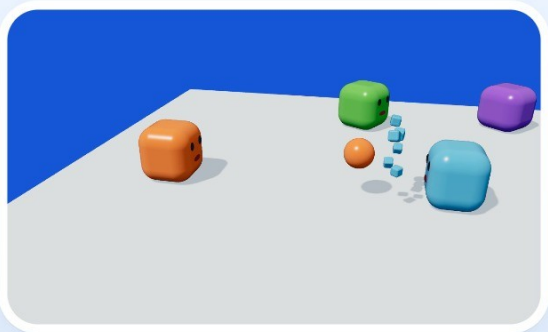
SPEED

Framework for after the MVP. Pips are relative; stats are not designed yet.

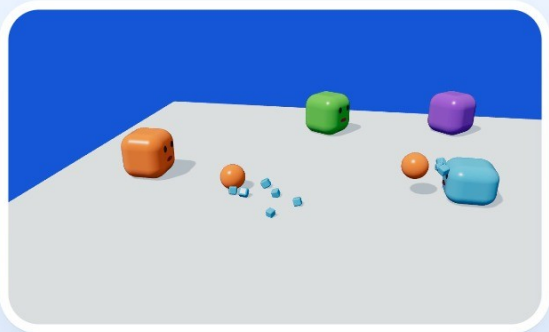
Prototype: we checked that the mass rule reads on screen



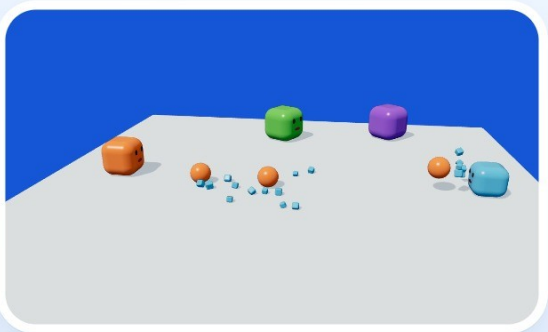
Before



Shot 1



Shot 2



Shot 3

WHAT WE BUILT

A deterministic physics simulation of the Mass Knockout rule in Three.js, rendered frame by frame.

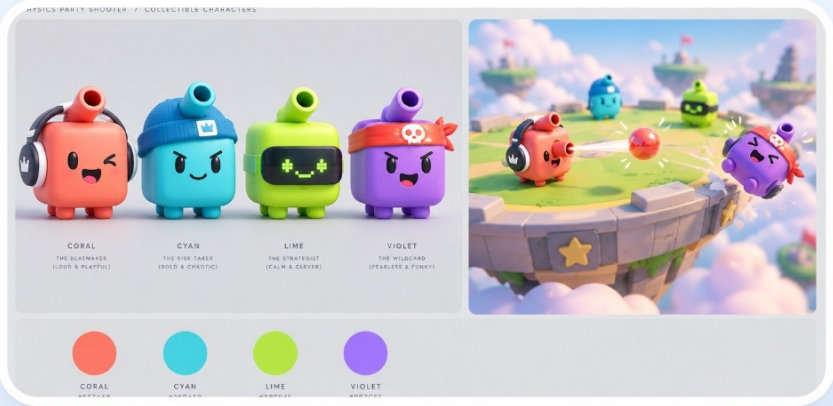
WHAT WE SAW

Over three shots the target shrinks and flies further each time. You can follow the rule without any UI.

HOW WE USE IT

It is the reference for our first Unity test of Mass Knockout in Demo 1: same scene, same rule.

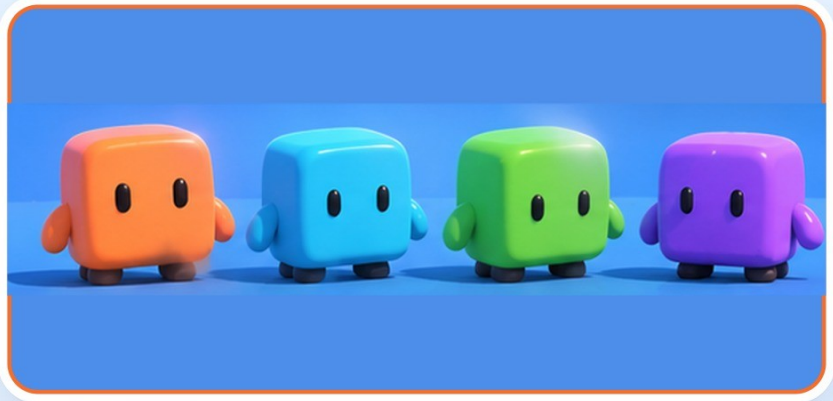
Four art directions for the same cube



A Vinyl toy



B Building blocks



C Glossy toy · chosen



D Jelly

PURPOSE

Four looks for the same character, generated to choose a direction before modelling.

USED IN THIS DECK

C, glossy toy. Every concept render in this deck uses it.

PLAYER COLOURS

Orange · Cyan · Lime · Purple

All AI-generated drafts. None is gameplay.

Logo drafts



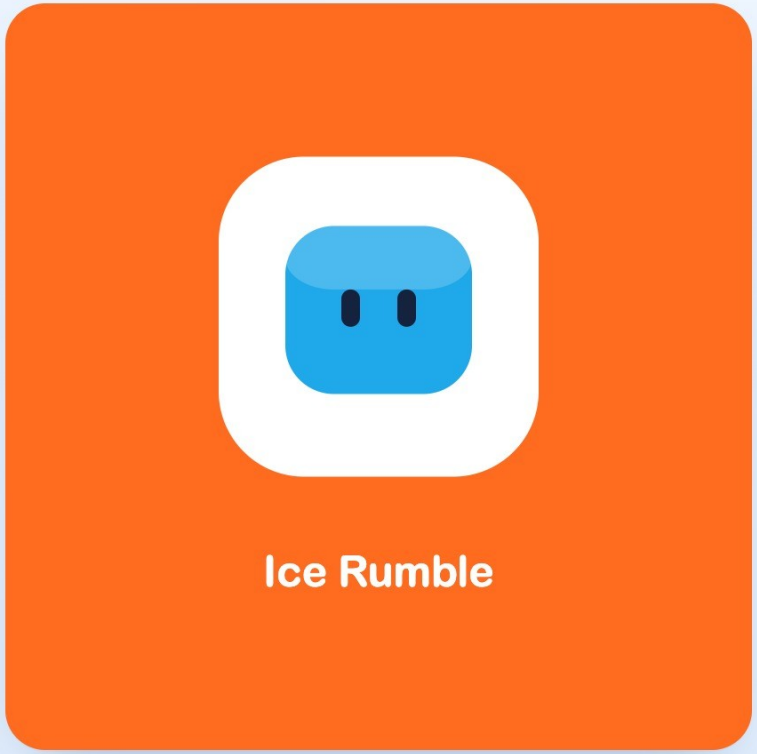
DRAFT A · WORDMARK

Title screen, deck cover, video end card.



DRAFT B · STACKED

Square formats: store capsule, lobby screen.



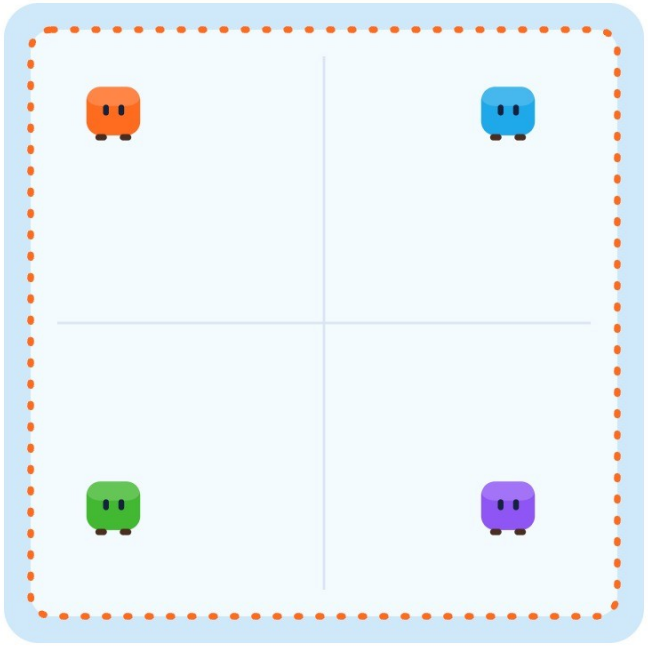
DRAFT C · ICON

Taskbar, build icon, favicon.

Drafts made for this proposal. The name is a working title; the logo is not final.

Level drafts, top-down: arena and track

MASS KNOCKOUT · ARENA DRAFT



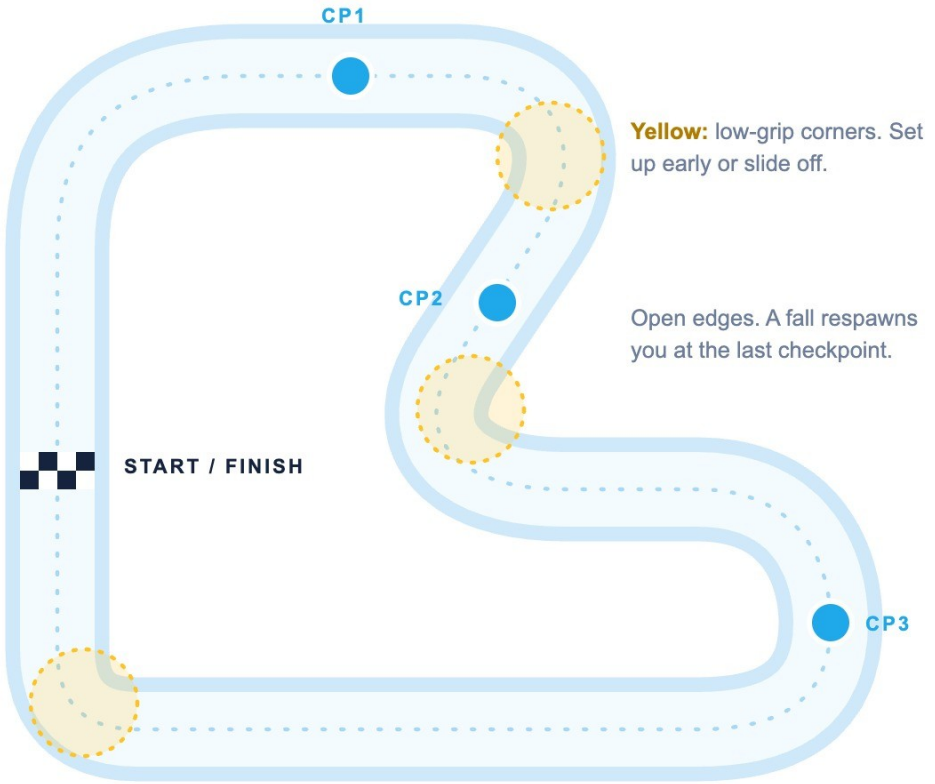
Fall zone. No walls, no cover.

One flat square platform.

Spawn near corners (draft).

OPEN EDGE = KNOCKOUT LINE

ICE RACE · TRACK DRAFT



Yellow: low-grip corners. Set up early or slide off.

Open edges. A fall respawns you at the last checkpoint.

Draft layouts for scope and blockout. Dimensions, spawn points and checkpoint count are not set.

Roadmap: four presentations, one playable step each

COURSE CHECKPOINTS · ALL MONDAYS

19 Oct

DONE

Proposal

Concept, research, prototype and drafts. This deck.

16 Nov

Demo 1

Shared core and LAN host/join. Ice Race playable as a blockout.

14 Dec

Demo 2

Mass Knockout over LAN. A full session with rounds and the scoreboard.

11 Jan

Final

Tuned, polished build. Promo video. Date subject to change.

RISK 01

Networked physics

Clients disagree on a knockback.

Host simulates; clients only render and interpolate.

RISK 02

Balance

Stalling, or a leader snowballing.

Score knockouts; full-mass respawn; timer; catch-up scoring.

RISK 03

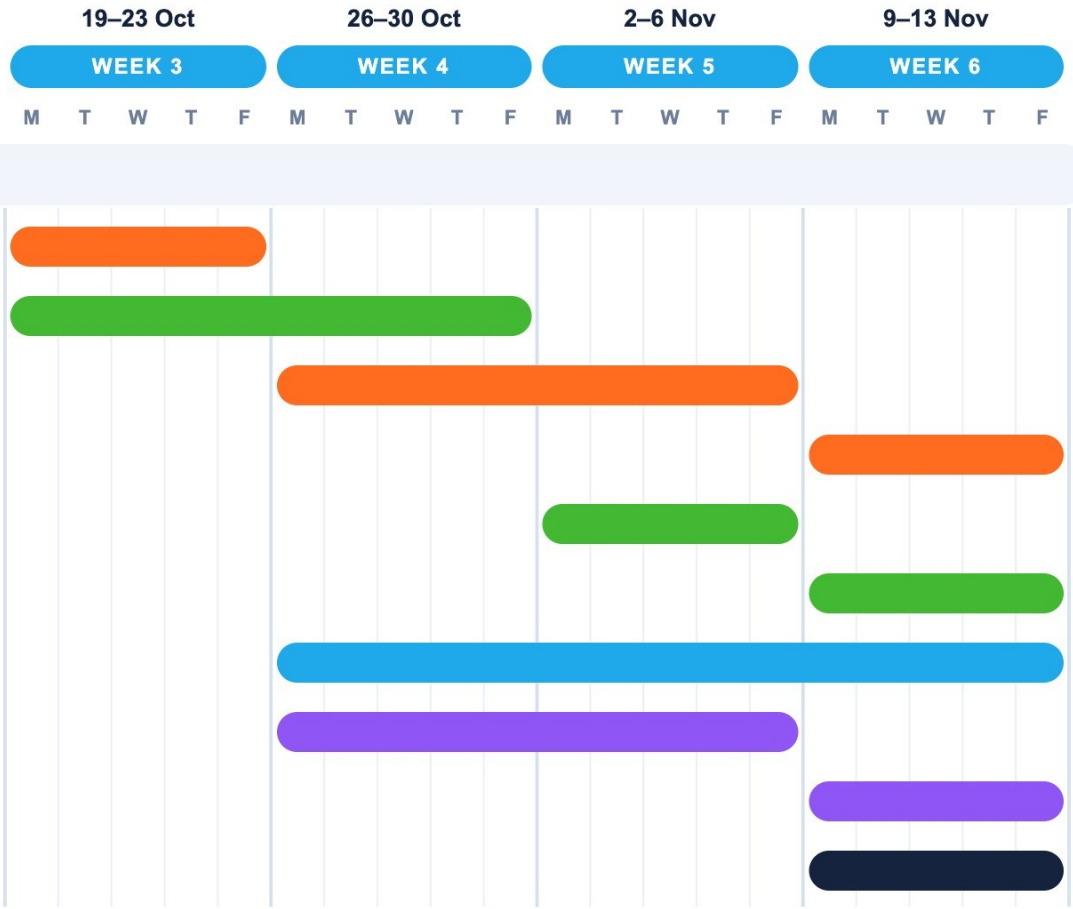
Scope

Too many mini-games for one term.

Exactly two in the MVP. More only once both are fun.

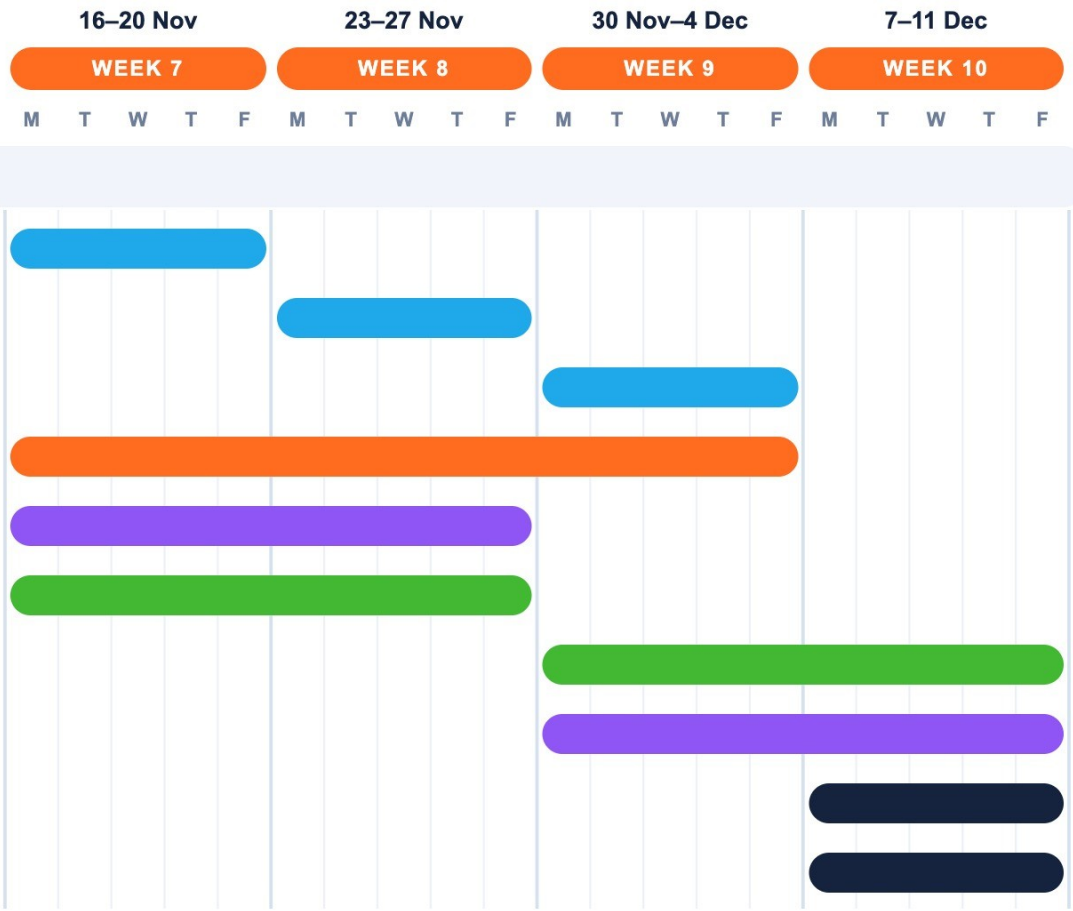
Gantt chart: demo 1, the shared core and Ice Race

WBS	TASK	OWNER	DURATION
2	**Demo 1 · 16 Nov**		
2.1	Unity project, scenes, build pipeline	Ezel	1 wk
2.2	Character controller & camera	Kürşat	2 wk
2.3	LAN host / join (Netcode)	Ezel	2 wk
2.4	Host authority & state sync	Ezel	1 wk
2.5	Ice movement profile	Kürşat	1 wk
2.6	Ice Race track blockout & checkpoints	Kürşat	1 wk
2.7	Mass state & spit projectile	Abdurrahim	3 wk
2.8	Lobby & join UI	Ardıl	2 wk
2.9	Character model, four colours	Ardıl	1 wk
2.10	Demo 1 build & presentation	Team	1 wk



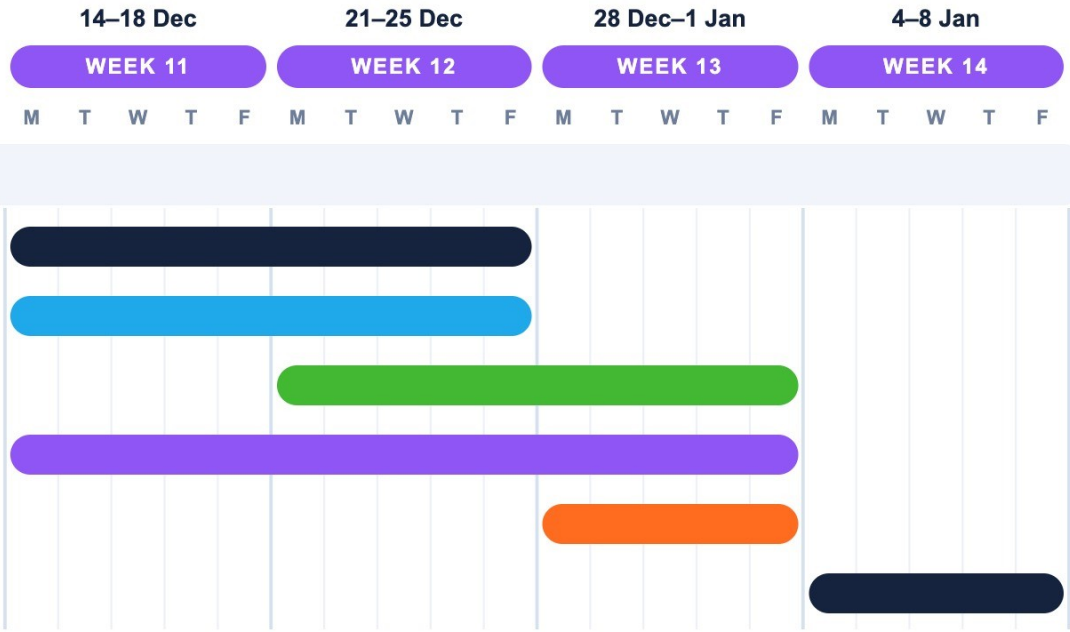
Gantt chart: demo 2, Mass Knockout and the full session

WBS	TASK	OWNER	DURATION
3	**Demo 2 · 14 Dec**		
3.1	Hit detection & mass loss	Abdurrahim	1 wk
3.2	Knockback scaled by mass	Abdurrahim	1 wk
3.3	Knockout credit & full-mass respawn	Abdurrahim	1 wk
3.4	Networked projectiles & hits	Ezel	3 wk
3.5	Mass Knockout arena	Ardıl	2 wk
3.6	Ice Race finish & placement	Kürşat	2 wk
3.7	Round state machine & rotation	Kürşat	2 wk
3.8	Scoreboard & catch-up scoring	Ardıl	2 wk
3.9	First full-session playtest	Team	1 wk
3.10	Demo 2 build & presentation	Team	1 wk



Gantt chart: final, tuning, polish and the promo video

WBS	TASK	OWNER	DURATION
4	**Final · 11 Jan**		
4.1	Balance tuning from playtests	Team	2 wk
4.2	VFX: trails, hit flash, splashes	Abdurrahim	2 wk
4.3	Sound effects & music	Kürşat	2 wk
4.4	Art & UI polish	Ardıl	3 wk
4.5	LAN stress test & final build	Ezel	1 wk
4.6	Promo video & final deck	Team	1 wk



PROPOSED OWNERS

- **Ezel** — Network, build
- **Abdurrahim** — Mass Knockout
- **Kürşat** — Movement, Ice Race
- **Ardıl** — Art, UI, session
- **Team** — Shared tasks

FINAL STANDINGS

Ice Rumble

1st 2–4 players, one LAN, no internet.

2nd Two mini-games on one shared Unity core.

3rd Every shot costs the shooter mass, so aim matters more than volume.



Concept render, AI-generated. Not gameplay.

Questions?

Ezel Bayraktar · Abdurrahim Özağaç · Kürşat Hürkan Aydın · Ardıl Bilal Ülkü

Sources

[1] Newzoo final 2025 figures, via GameDev Reports, 2026-06-26. gamedevreports.substack.com/p/newzoo-gaming-market-surpassed-200

[2] Alinea Analytics, games with co-op on Steam, H1 2025 (proprietary estimates). alineaanalytics.substack.com/p/games-with-co-op-generated-over-4

[3] ESA Essential Facts 2026 (survey Feb 2026, n=13,545). theesa.com

[4] Steam store search, tag “Party Game” × LAN PvP / Online PvP / Shared-Split-screen PvP, 2026-10-17. Tags are user-applied.

[5] Game Developers, 2025 Steam revenue distribution (Gamalytic estimates), 2026-08-21. game-developers.org/2025-steam-game-revenue-distribution

[6] PC Gamer, Valve at GDC 2026: 5,863 games over $100K on Steam in 2025. pcgamer.com

[7] Steam store pages and SteamDB charts, apps 1260320, 285900, 1097150, 880940, 1858400: prices, peaks, reviews, owner estimates. 2026-10-17.

[8] GameRevolution / Prima Games: Party Animals requires an internet connection, including split-screen.

[9] Osmos official site, osmos-game.com. [10] Squarena on Steam (app 1858400); codeaclysmgames.com/squarena-details.

[11] Morphies Law official site, morphieslaw.com/game. [12] Crisol: Theater of Idols, Blumhouse Games and Steam (app 1790930).

Images: all game visuals are AI concept renders (GPT Image via Higgsfield; MiniMax H3) from our own reference art, except competitor screenshots (official Steam media). Logo and level drafts made for this proposal.